



Figure 4: VTN OpenStack architecture

Installing OpenDaylight

The OpenDaylight controller runs in a JVM. The OpenDaylight-Boron release requires OpenJDK8, which you can install using the command given below:

```
sapt-get install openjdk-8-jdk
```

Download the latest OpenDaylight-Boron package from the official repo, as follows:

```
$wget https://nexus.opendaylight.org/content/repositories/
opendaylight.release/org.opendaylight/integration/
distribution-karaf/0.5.1-Boron-SR1/distribution-karaf-0.5.1-
Boron-SR1.tar.gz
```

Untar the file as the root user, and start OpenDaylight using the commands given below:

```
$ tar -xvf distribution-karaf-0.5.1-Boron.tar.gz
$ cd distribution-karaf-0.5.1-Boron.tar.gz
$ ./bin/karaf
```

Now, you should be in OpenDaylight's console. Install all the required features, as follows:

```
opendaylight-user@root> feature:install odl-vtn-manager-
neutron
opendaylight-user@root> feature:install odl-vtn-manager-rest
opendaylight-user@root> feature:install odl-mdsal-apidocs
opendaylight-user@root> feature:install odl-dlux-all
```

Feature installation may take some time. Once the installation is complete, you can check whether everything is working fine by using the following `curl` call:

```
$ curl -u admin:admin http://<ODL_IP>:8080/controller/nb/v2/
neutron/networks
```

The response should be an empty network list if OpenDaylight is working properly.

Now, you should be able to log into the DLUX interface

on `http://<ODL_IP>:8181/index.html`.

The default username and password are `admin/admin`.

Additionally, you could find useful log details at the following location:

```
$ tail -f <directory_of_odl>/data/log/karaf.log
$ tail -f <directory_of_odl>/logs/web_access_log_2015-12.txt
```

Now, you have a working OpenDaylight-Boron set-up. Let's get into the integration part.

Configuring OpenStack for VTN integration

Step 1

Erase all VMs, networks, routers and ports in the controller node, since you already have a working OpenStack set-up. You might test for VM provisioning as a sanity test, but before integrating OpenStack with OpenDaylight, you must clean up all the unwanted data from the OpenStack database. When using OpenDaylight as the Neutron back-end, ODL expects to be the only source for Open vSwitch configuration. Because of this, it is necessary to remove existing OpenStack and Open vSwitch settings to give OpenDaylight a clean slate.

The following steps will guide you through the cleaning process.

- Delete instances, as follows:

```
$ nova list
$ nova delete <instance names>
```

- Remove links from subnets to routers, as follows:

```
$ neutron subnet-list
$ neutron router-list
$ neutron router-port-list <router name>
$ neutron router-interface-delete <router name> <subnet ID or name>
```

- Delete subnets, nets and routers, as follows:

```
$ neutron subnet-delete <subnet name>
$ neutron net-list
$ neutron net-delete <net name>
$ neutron router-delete <router name>
```

- Check that all ports have been cleared – at this point, this should be an empty list:

```
$ neutron port-list
```

- Stop the Neutron service, as follows:

```
$ service neutron-server stop
```

While Neutron is managing the OVS instances on the compute and control nodes, OpenDaylight and Neutron may be in conflict. To